PROmpter — Manual

PROmpter automatically turns the export files of the Roche **LightCycler PRO** into a readable **PDF report** and an **Excel spreadsheet** (for later LIS processing).

> **Important:** PROmpter is a pure **presentation/reporting tool**. All values (Cq, finalCall, curves) come **unchanged from the LightCycler PRO**. PROmpter does not recalculate anything and does not reinterpret anything — it merely summarises the device's own results clearly. The reports are an **interim evaluation**; the professional sign-off is still done manually. PROmpter is **initially intended for qualitative analyses** — i.e. the **reactive / non-reactive** call per target, as the LightCycler PRO provides it in `finalCall`. **Quantitative** evaluations (absolute or relative quantification via standard curves) are **currently not** supported.

In a nutshell

1. In the **Settings**, choose a **watched folder** (where the LC PRO drops its export files).

2. **Start monitoring** — done. Every new file is processed automatically.

3. For each file a **PDF** and an **Excel** are created; the **original** is then moved to the archive (or deleted — depending on the setting).

PROmpter may run in the background: with **"Work in background"** it sits as an icon in the notification area (tray) next to the clock, and monitoring stays active. The **window's close button (), in contrast, quits PROmpter completely.**

Installation & first start

PROmpter does **not** need to be installed — it is a single `.exe`. It is best to copy it to a **fixed local location** (e.g. `C:\PROmpter\`) and start it by double-clicking; a desktop shortcut makes it convenient. **Administrator rights are not required** — neither on the first nor on later starts.

On the very first start Windows may show a blue message **"Windows protected your PC"** (Microsoft Defender SmartScreen). This is **not** an error and **not** a virus, just a notice because the program is not (yet) digitally signed. Here is how to continue:

1. Click **"More info"**.

2. Then click **"Run anyway"**.

This is only necessary **once per program version** — after that PROmpter starts without asking. (If the file is in a OneDrive folder, the message may reappear; at a local location such as `C:\PROmpter\` it stays a one-time thing.)

Note for clinical/laboratory IT

PROmpter is a **stand-alone, portable** Windows application (Python/PySide6/Qt) **without an installer, without administrator rights and without a network service**. It reads only the configured **watched folder** and writes **PDF/Excel** to the chosen output folders, plus its settings to `%APPDATA%\PROmpter\`. The **only** optional outgoing network access is a **version check** when clicking "" (fetching a small text file; can be disabled by leaving out the update URL).

So that staff do not see the SmartScreen message at all, IT can **whitelist** the file — e.g. via a SmartScreen exception or through AppLocker/WDAC. The cleanest long-term solution is to **sign** the `.exe` with a **code-signing certificate** of the institution; then the warning disappears entirely.

Exporting the run at the LightCycler PRO

For PROmpter to process a run, it must be exported at the LightCycler PRO as **customer data** (a ZIP file with XML). Here is how at the device:

1. Open the **Results** application and tick the **completed run** in the **run overview**.

2. Choose the **Export** button.

3. In the **"Select location for file and export"** dialog, choose the option **"Export customer data"**.

> **Important:** do *not* choose "Perform data export" — that creates an **LCRD** file (folder `run_data`) for the Roche development software. That file is **AES-encrypted** (Roche's protected internal format) and **cannot** be read by PROmpter at all. PROmpter needs the open **customer data** (ZIP/XML).

4. Choose the **location** (SFTP server or USB device) and confirm with **Export**.

The customer data is stored as a **ZIP archive** (named after the device serial number) in the `customer_data` folder; inside it there is one XML per run (named after run ID + timestamp). You set **exactly this target folder** (or the folder into which the SFTP transfer lands) as the **watched folder** in PROmpter — then every new export is automatically processed into PDF + Excel.

> **Tip:** Several completed runs can also be **exported together** — PROmpter then creates a separate report for **each run** (see "Several runs in one file").

The main window

- **Start/Stop monitoring** — switches the automatic folder monitoring on/off.
- **Process file now …** — select and process a single export file by hand.
- **Scan folder now** — search the watched folder once for files not yet processed.
- **Settings …** — all options (see below).
- **Work in background** — places the window into the notification area (tray); monitoring keeps running in the background (bring it back via the tray icon's "Show", "Quit" closes it completely).
- **Close** — the **window's close button ()** quits PROmpter **completely** (not to the notification area).
- **Log** — shows for each file the time, name, status (**green** = OK, **yellow** = notice/partially processed, **red** = error) and details. If something fails, **the original file stays put** — nothing is lost.
- **Save log …** — save the displayed lines as **CSV** (for Excel) or as a **text file**.
- **Clear log** — clears only the display; the PDF/Excel files already created are kept.

Settings

The settings are organised in three tabs: **File management** (folders, which files, original action, printing, file names), **PDF report** (sections, header/footer, sorting, curve type) and **Excel report** (columns, sorting, sheets, curve type). **Sorting and curve type can be set separately per report** — the PDF may e.g. be by column and the Excel by row.

For **sorting the samples** there are four options: **by column** (A1, B1, C1 …), **by row** (A1, A2, A3 …) as well as **alphabetically ascending or descending** by the **sample name**.

Sprache / Language

At the very top of the **File management** tab, the **language** can be changed: **Deutsch** (default) or **English**. The change affects the **interface, the PDF report, the Excel file and all messages**; in English mode numbers are written with a **point** instead of a comma (e.g. Cq `15.62`). **Device and sample data** (assay/sample names, Cq, finalCall, curves) stay unchanged in **both** languages. The new language takes effect **after a restart** of the program. *(The manual is available in German and English and opens automatically in the language that is set.)*

Folders

- **Watched folder** — is monitored automatically.
- **PDF output / Excel output** — target folders. **Empty = same folder as the export file** (the finished PDF/Excel are not read in again).
- **Archive folder** — where the original is moved to. Empty = subfolder `Archiv/`. (Deliberately a **separate** folder so that the moved original is not immediately processed again as a "new file".)

Original file after processing

- **Move to archive** (default) or **delete**. *"Leave in place" deliberately does not exist*, so that the same run is not accidentally processed twice.

Printing

PROmpter can **automatically print the PDF report after it has been created** — handy for the paper archive of an unattended watched folder.

- The **checkbox "Print PDF report automatically after creation"** turns the function on (default: off).
- **Printer** — the **default printer** (Windows) or a specific printer from the list of installed printers. This lets the monitoring PC print specifically to the **lab-report printer**.
- **Copies** — number of printouts per report (e.g. 2 for a duplicate archive).
- **PROmpter prints the PDF itself** directly to the chosen printer — **regardless of which PDF program** is installed or set as default on the PC (no Adobe/Edge needed).
- **Safety: only the generated PDF** is ever printed, never the original file. If printing fails once (e.g. no printer reachable), the run is still **counted as processed** (PDF/Excel are saved after all) and marked **yellow** with a notice in the log.

Sorting the samples

- **by column** (A1, B1, C1 …) or **by row** (A1, A2, A3 …) — selectable **separately** in the *PDF report* and *Excel report* tabs.
- **Controls (NTC/PTC)** always appear in the Excel, clearly marked as a control. In the **summary** (Excel) and the **Cq table** (PDF) they are additionally highlighted: **PTC red**, **NTC green**, **reactive samples bold**.

Curve display

Which curve form computed by the device is displayed (selectable **separately** in the *PDF report* and *Excel report* tabs). In brackets the name that the **LightCycler PRO** itself uses for it:

Selection (PROmpter)	At the device	Meaning
Baseline divided (default)	"Baseline divided" (`bdcurve`)	Each curve point is **divided by the baseline**. **This is what the LC PRO always computes its results from** — hence the most meaningful display.
Baseline subtracted	"Baseline subtracted" (`bscurve`)	The **baseline is subtracted from each point**.
Raw values	"Raw data" / "Fluorescence history"	The **unprocessed** measured fluorescence.

The **baseline** is the y-axis intercept of an amplification curve; each curve has its own. All three forms come **unchanged from the device** — PROmpter does not compute its own baseline. *(Terms as per the LightCycler PRO user assistance.)*

Two curve views (each individually switchable on/off in PDF and Excel):

- **per sample** — a small chart per well with all channels (overview per patient sample); the targets/channels are coloured as a **spectrum blue -> dark red**.
- **per target** — *one* chart per channel/target, in which **all samples lie on top of each other** (the classic amplification-curve view). The curves are **coloured by result** (reactive coloured, negative grey, controls highlighted); reactive curves and controls are **numbered**, with a small **mapping table** next to them (no. -> well -> sample -> Cq). In Excel, each sample is additionally its own line whose name appears **on mouse-over**. The **Y scale is aligned per target to the strongest sample** (the fluorescence level differs from target to target).

File names of the reports

The name of the generated PDF/Excel is assembled from **building blocks**. Buttons insert placeholders; the **preview** immediately shows the result. *(The placeholders are inserted by buttons and are the same in both languages.)*

Placeholder	Content
`{Plattenname}`	Name of the plate setup, e.g. "260609_BovPev_Myco"
`{Datum}`	Run date, e.g. 2026-03-27
`{DatumZeit}`	Date + time
`{Uhrzeit}`	Time
`{LaufID}`	Run ID
`{PlateID}`	Plate ID
`{LCAP}`	1st LCAP of the run, e.g. "GastroV96" (the old name `{Assay}` still works)
`{Profil}`	PCR profile, e.g. "mPCR7"
`{LCAPVersion}`	Version of the 1st LCAP, e.g. "1.0" (the old name `{AssayVersion}` still works)
`{Geraet}`	Device serial number
`{Bearbeiter}`	Operator from the file

Free text is allowed (e.g. `Result_{Datum}_{LaufID}`). Empty blocks and invalid characters are cleaned up automatically; if the name is the same, PROmpter appends _2, _3 … and **never overwrites**.

Header and footer of the PDF

- **Header** — up to **two freely chosen lines** (e.g. "Example University" / "Virus diagnostics"). Below it the **plate name - run ID** is shown.
- **Footer (left)** — freely composed from **building blocks** (the same placeholders as for the file name), default `{Datum}_{PlateID}_{LaufID}`. The **page number** stays on the right.

PDF — which sections appear

Header, control-validity check, plate overview, result overview, individual values (Cq), **amplification curves per sample** and **amplification curves per target** — each section individually switchable on/off.

Excel — which sheets appear

As with the PDF sections, a checkbox can be set per **sheet**: **Result table**, **Summary**, **Amplification curves per sample**, **Amplification curves per target**, **Reactive samples**.

Excel — which columns appear

A checkbox per column (for the "Results" sheet). The table is built "long" (one row per sample × target) — good for the later LIS import.

The outputs

PDF report

Header with run key data · control-validity check (NTC/PTC) · coloured **plate overview (96 or 384 wells, automatically detected)** · **result overview per sample** (overall statement, reactive targets with Cq, internal control) · **individual values** (Cq per target) · **amplification curves per sample** (one chart per well, all channels) · **amplification curves per target** (all samples per channel overlaid, coloured by result + mapping table). Uniform Y scale **from 0**, aligned to the samples.

Excel workbook

- **Results** — the configured columns (flags in plain text). The **"Flags"** column shows **all** flags, the separately selectable **"Critical flags"** column only the heavily weighted ones.
- **Summary** — cross table: one row per sample, one column per detection target with the Cq (**empty = not tested, "–" = tested but negative**), plus internal control, overall result and the **"Critical flags"** column (plain-text reason), plus a tally.
- **Curves** — native, editable Excel line charts per sample.
- **Curves per target** — *one* chart per channel with all samples (coloured by result; sample name on mouse-over).
- **Reactive samples** — the same per-sample charts, only for reactive samples.

How PROmpter summarises (transparently)

- **reactive** — as soon as **at least one** target is reported by the device as reactive/positive (the affected targets are named).
- **non-reactive** — all detection targets are non-reactive/negative.
- **see individual values** — mixed or unexpected (e.g. an "Invalid").
- **Internal control (IC)** — is only **shown**; it does **not** automatically block the result.
- **Flags** are shown in **plain text** (e.g. "EPF value too low").

This overall statement is **not a new assessment**, only a bundling of the device's own individual results.

Several runs in one file

If the LightCycler PRO exports **several runs into one file** (one ZIP archive, named after the device serial number), PROmpter creates a separate PDF and a separate Excel for **each run** (named after the respective run). This works **reliably** — so you can safely export several completed runs **together**.

Background: each run is a **self-contained, complete XML file** in the ZIP (named after run ID + timestamp); PROmpter reads them one after another. Should a single run exceptionally be **unreadable** once (e.g. a corrupted export file), PROmpter still processes the remaining runs and marks the file **yellow** in the log with a notice — the whole file is never discarded, and the original is, in this case, **archived instead of deleted** to be safe.

Notes & limitations

- PROmpter is initially designed for **qualitative analyses** (reactive / non-reactive per target). **Quantitative** evaluations (quantification via standard curves) are **currently not** supported.
- PROmpter is a **helper tool** and **not validated**. Please check the result, controls and plausibility before every sign-off.
- The export files carry `sentToHost=false` / status "unreviewed" — so they are **not yet released**.

- PROmpter **never** writes to the original data; it only reads and creates new files.

Version history

Version 1.8 (June 2026)

- **English language variant:** PROmpter now speaks **German and English** — switchable in the **Settings** (*File management -> "Sprache / Language"*, default German). The change affects the interface, PDF report, Excel and all messages and takes effect after a restart; in English mode numbers are written with a **point** instead of a comma. **Device and sample data stay unchanged in both languages** (PROmpter does not recalculate or reinterpret anything). The manual is now available in German and English and opens automatically in the language that is set.

Version 1.7 (June 2026)

- **Folder monitoring on network drives:** the automation now actively polls the watched folder at a fixed interval (*polling*) instead of waiting for operating-system events. This means new exports are reliably detected even on **network/SMB shares** (where the native file events failed to fire — especially when another computer is writing). The polling interval can be set in the **Settings** (*File management -> "Poll folder every … sec."*, default 5 s); only the directory is polled, not the file contents (minimal load).
- **Excel — critical flags in plain text:** new, deselectable **"Critical flags"** column in the *Results* sheet (next to "Flags", which still shows **all** flags) and a new **"Critical flags"** column at the end of the **Summary** — as in the PDF overview, only the heavily weighted flags.
- **Summary like the PDF overview:** in the cross table an **empty cell means "not tested"** and **"–" means "tested but negative"** (previously both empty and thus indistinguishable).
- **"LCAPs & assays" block clearer:** per target now **"Name (dye excitation/emission)"**, e.g. "Parechovirus (FAM 494/523)" — dye and channels in *one* bracket. In addition the **internal control** is now listed too — appended at the end and marked with **"IC:"** (e.g. "…, IC: Internal control (Cy5 657/675)").
- **Excel — channel & dye:** the **"Channel"** column now additionally shows the excitation/emission wavelengths (**"22 (494/523)"**), and there is a new, deselectable **"Dye"** column (e.g. FAM, R6G, Cy5).
- **File-name preview:** the preview in the settings dialog now shows the `{Plattenname}` block with an example value (it was always inserted correctly in the real file, but remained empty in the preview).

Version 1.6 (June 2026)

- **Automatic printing:** the PDF report can be printed directly after creation — with **printer selection** (default or specific printer) and **number of copies**. Configurable in the *File management -> Printing* tab. PROmpter prints the PDF **itself** (independent of the installed PDF program).
- **New header block** in the PDF: **plate name** (from the plate setup), **PCR profile**, **reaction volume**, **cycles**, **device incl. block** (e.g. "96 block"), **plate ID**, **block serial no.** and **instrument software**. The subtitle line shows **plate name - run ID**.
- **Plate name** as a new file-name block `{Plattenname}` and as the (first) **Excel column**.
- In the "LCAPs & assays" block, each target now additionally shows the **dye** (e.g. "FAM"), not just the excitation/emission channels.
- **"Assay" -> "LCAP" clarified:** the Excel column is now called **"LCAP"** and shows **each sample's own LCAP** (previously the first LCAP of the run everywhere). The file-name block `{Assay}` is now called `{LCAP}` (or `{LCAPVersion}`); the old names still work.
- **Reason for "invalid" visible:** for invalid samples the **critical device flags are shown in plain text** — in the **result overview** (column "Targets", e.g. "Validation failed: invalid internal control; Invalid result") and in **"amplification curves per target"** as a **numbered footnote** directly below the mapping table. (Only critical flags; pure info flags are not listed.)

- **384 plates:** the **plate overview detects the format automatically** (96 = 8×12, 384 = 16×24) and draws the matching grid — in portrait. The remaining evaluations (overview, Cq, Excel, curves) were already format-independent anyway.

Version 1.5 (June 2026)

- This version overview added to the manual.

Version 1.4

- The "Individual values — Cq per target" table is shown **automatically in landscape** when there are **many targets** (otherwise portrait).
- **Empty cell = target not in the sample's panel**; **"–" = tested but negative** (no Cq).

Version 1.3

- **Channels (excitation/emission)** behind each target, e.g. (494/523).
- Better distinguishable, **colour-blindness-friendly curve colours**.
- Alternately shaded table rows for better readability.
- Target order **grouped by LCAP**.
- New **"Result"** column (reactive / invalid / control) in "amplification curves per target".
- Excel: reactive **and invalid** results in bold; "Type" column removed from the summary.

Version 1.2

- Support for runs with **several LCAPs (assays)**: all LCAPs including targets in the header, each target as its own column.
- New result category **"invalid"** (clearly marked).
- Fix: negative results are reliably recognised as **"non-reactive"**.

Version 1.1

- Bug fix in PDF generation with very many reactive samples on one target.

Version 1.0

- First version: automatic processing of the LightCycler PRO customer data into a **PDF report** and an **Excel spreadsheet**, including amplification curves, configurable reports and an update check.

License & open-source libraries

- You **may** pass PROmpter on to colleagues and use it in the lab. Private/free distribution is unproblematic.
- PROmpter uses open-source libraries (no part of Roche's own):
- **PySide6/Qt** (© The Qt Company) — license **GNU LGPL v3** · <https://www.qt.io/licensing>
- **ReportLab** (PDF generation) — **BSD license** · <https://www.reportlab.com>
- **openpyxl** (Excel generation) — **MIT license** · <https://openpyxl.readthedocs.io>
- **watchdog** (folder monitoring) — **Apache 2.0 license** · <https://github.com/gorakhargosh/watchdog>

These licenses permit the use in PROmpter without disclosing your own source code. As long as the app is passed on **as an unchanged, stand-alone `.exe`**, all license obligations are met. (You will find the same note in the **/About** dialog of the app.)

- **Roche** and **LightCycler** are trademarks of their respective owners. PROmpter is an **independent** tool, **not** reviewed or supported by Roche.

Feedback

Suggestions for improvement and bug reports are welcome — via the **button** in PROmpter (e-mail or online form).